AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION 2010

Physics Paper I

Time allowed: 35 minutes Marks 25

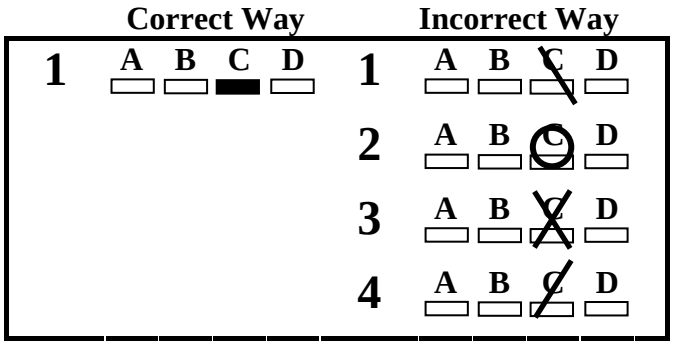
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

7. You may use a simple calculator if you wish.

1. If a radio station broadcasts its transmission at 600 kHz, with a speed of radio waves at 3×10^8 m/s, then the wave length is

   A. 300 m
   B. 400 m
   C. 500 m
   D. 600 m

2. If a stone is dropped in a tank of water, then the waves formed on the surface of the water are called

   A. travelling waves.
   B. stationary waves.
   C. transverse waves.
   D. longitudinal waves.

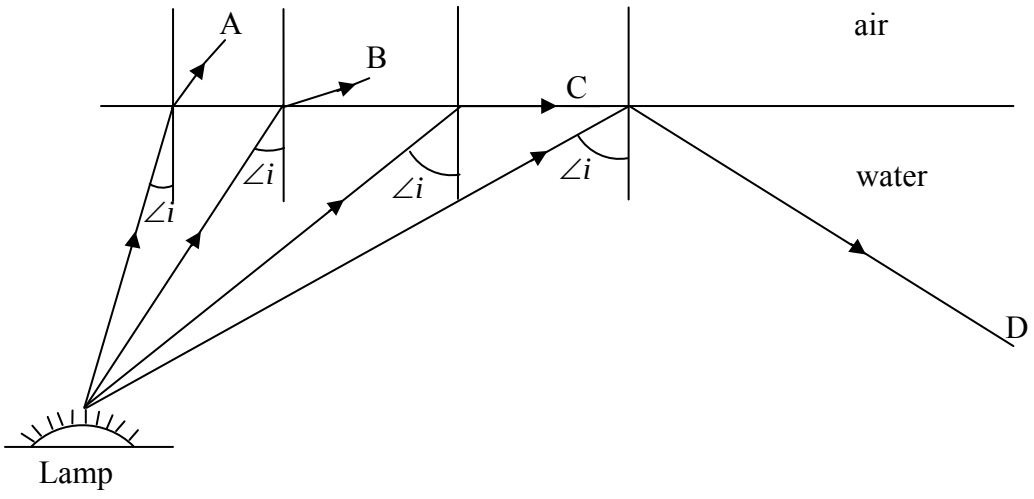
3. Waves transmit

   A. force.
   B. inertia.
   C. energy.
   D. momentum.

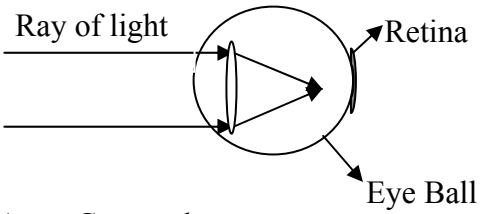
4. We can distinguish between two sounds of the same loudness and pitch by

   A. the audible range.
   B. the quality of sound.
   C. different instruments.
   D. the sensitivity of the ear.

5. Which ray of light makes a critical angle?

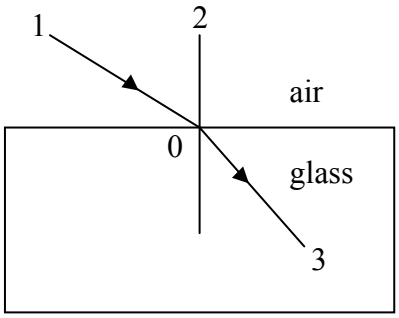


6. A defect of a human eye is shown in the diagram. Which optical media will be used to correct this defect?



   A. Convex lens.
   B. Concave lens.
   C. Convex mirror.
   D. Concave mirror.

7. Which is the correct representation of rays shown in the diagram?



	1	2	3
A.	Incident ray	Refracted ray	Normal
B.	Refracted ray	Normal	Incident ray
C.	Normal	Refracted ray	Incident ray
D.	Incident ray	Normal	Refracted ray

8. An object of 3 cm height is lying at a distance of 9 cm from a concave mirror. If the image is formed at a distance of 12 cm from the pole of the concave mirror, then the size of the image is

   A. 4 cm
   B. 9 cm
   C. 15 cm
   D. 36 cm

9. Materials with a high resistance to the flow of current and capable of being polarized by an electrical field are

   A. insulators.
   B. dielectrics.
   C. conductors.
   D. semiconductors.

10. A capacitor is a device used to

    A. store charge.
    B. convert A.C to D.C.
    C. stop the flow of current.
    D. stop the function in the electric supply.

11. All of the following are properties of electric lines of force EXCEPT

    A. they all cross each other.
    B. they are vector quantities.
    C. they define the intensity of electric fields.
    D. they start from positive charge and terminate at negative charge.

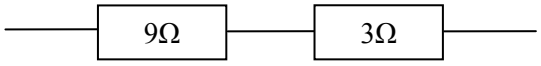
12. A device which maintains current and voltage for different components in an electric circuit is known as

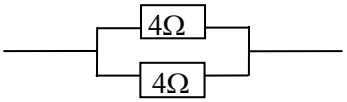
    A. a battery.
    B. a voltmeter.
    C. an ammeter.
    D. a transistors.

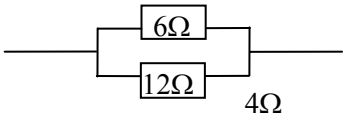
13. Which of the following fuses would be the most suitable for a circuit in which a 2000 W kettle is connected to a 240 V supply?

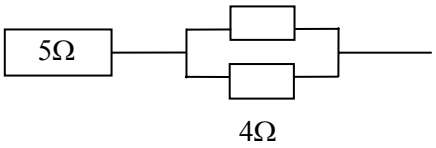
    A. 3 A
    B. 6 A
    C. 8 A
    D. 12 A

14. Which one of the following circuits gives the combined resistance of 2Ω?

A 

B 

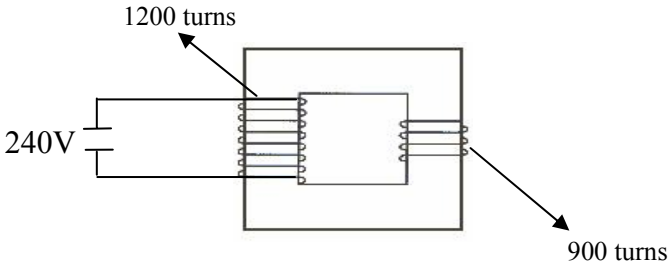
C. 

D. 

15. An electric motor converts one form of energy to another. Which of the following conversions of energy is applicable to the above statement?

    A. Electrical to heat.
    B. Electrical to solar.
    C. Electrical to mechanical.
    D. Electrical to geo-thermal.

16. Using the information in the given figure, the secondary voltage will be



A. 180 V
B. 240 V
C. 320 V
D. 4500 V

17. The number of protons in a nucleus is called

A. the mass number.
B. the atomic number.
C. the neutron number.
D. Avogadro's number.

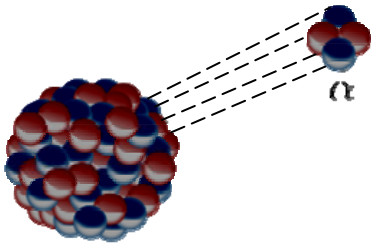
18. The emission of rays from the nucleus is called

A. radioactivity.
B. atomic process.
C. chemical process.
D. atomic dispersion.

19. A reaction which occurs on the surface of the Sun is

A. fusion.
B. fission.
C. annihilation.
D. pair production.

20. During alpha decay which parts of the atom decrease in number?

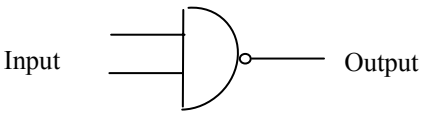


A. Protons and neutrons.
B. Electrons and protons.
C. Electrons and neutrons.
D. Electrons, protons and neutrons.

21. A diode is a two terminal device which allows current to flow in

A. one direction.
B. two directions.
C. many directions.
D. reverse direction.

22. The name of the given electronic gate is



A. OR
B. NOT
C. AND
D. NAND

23. Carbon semi-conductor diode is NOT used in electronics because

A. it is an insulator.
B. it has low potential.
C. it has high potential.
D. it converts AC into DC.

24. Which is penta-valent impurity?

A. Silver
B. Indium
C. Arsenic
D. Gallium

25. Devices which control the function of electronic machines without any cable connection are known collectively as

A. Joy sticks.
B. Radio controls.
C. Remote controls.
D. Computer mouses.

END OF PAPER